

The VSOP Manual

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Contents

1	Introduction	1
2	Lexical Structure	2
2.1	Whitespace	2
2.2	Comments	3
2.3	Integer Literals	3
2.4	Keywords	3
2.5	Type Identifiers	3
2.6	Object Identifiers	4
2.7	String Literals	4
2.8	Operators	4
3	Syntax	5
4	Semantics	5
4.1	Classes	7
	New	7
	Inheritance	7
	Dispatch	8
	Fields	8
	Methods	9
	The Object Class	9
	The Main Class	9
4.2	Expressions	9
	Literals	9
	Identifiers	10
	Assignments	10
	Dispatch	10
	Conditionals	11
	Loops	11
	Blocks	11
	Let	12
	Arithmetic, Logic and Comparison Operations	12
5	Suggested Extensions	12
5.1	Add New Control-Flow Structures	12
5.2	Indentation-based Syntax	12
5.3	Easier Let-In Constructs	13
5.4	Add other numeric types	13
5.5	Add Missing Operators	13
5.6	Add a Foreign Function Interface (FFI)	13

5.7 Dispatch on Parent Class	14
5.8 Addition of Arrays	14
5.9 Local Type Inference	15
5.10 Top-level Functions and/or Static Methods	15
5.11 Add Static Fields	15
5.12 Multiple Inheritance or Interfaces	15
5.13 Generics/Templates	15
5.14 Visibility	15
5.15 Default and Named Arguments	16
5.16 Add Support for Multiple Return Values (Anonymous Structs)	16
5.17 Add an Error Handling Mechanism	16
5.18 Separate Compilation	17
5.19 Garbage Collection	17
A Examples	18
A.1 Factorial	18
A.2 Linked List	18

1 Introduction

The *Very Simple Object-oriented Programming* language (or VSOP) is, as its name implies, a simple object-oriented language. It is simple enough that a compiler for it can be developed in a short amount of time, but still retains some features of modern programming languages. VSOP is inspired from COOL, the *Classroom Object-Oriented Language* by Alex AIKEN *et al.*

Here are the more salient features of VSOP. Some of them will be explained in more details later in this document, some during the lectures.

- VSOP is meant to be a *general-purpose* programming language. *I.e.* it is not specialized to a specific domain like $\text{\TeX}$ or Prolog, but should allow to write various kinds of programs.
- With features such as automatic memory management and class-based dispatch, VSOP is a somewhat *high-level* language.
- VSOP is an *object-oriented* language. Programs are structured as a set of user-defined *classes*. Each class specifies both a data structure by the way of private member *fields*, and operations on that data structure, by the way of member *methods*. This provides information hiding and encapsulation.

VSOP also supports single-parent *inheritance*.

- VSOP is an *expression-based* language. Nearly all constructs in VSOP are *expressions*, with a type and a value. *E.g.* the expression

```
if cond then e1 else e2
```

has a type and a value, which will be equal to the value of `e1` or the value of `e2`, depending on the runtime value of `cond`. You could then write, *e.g.*

```
someFunction(if cond then e1 else e2)
```

It is thus more similar to C's ternary operator `cond ? e1 : e2` than to C's *if-then-else* construct.

Most other VSOP constructions are also expressions.

- VSOP is *statically typed*, *i.e.* the compiler checks the types of expressions at *compile-time*. This ensures that there will be no type errors at runtime, and allows efficient code generation.

It is also somewhat *strongly typed*, by providing no mechanisms to work around the type system, like unsafe casts.

- VSOP is *explicitly typed*, i.e. the developer must annotate its code with typing information. While it is certainly a burden, it can help to make programs more legible, and allows the compiler to do a simple *type checking* instead of a much more involved *type inference*.
- VSOP is a *sequential* language. It has no provision for *concurrent* programming, although multi-threaded programs are possible through library extensions.
- VSOP supports automatic memory management through *garbage collection*.
- VSOP has no support for error handling, like for example an *exception* mechanism.
- VSOP is a *strict*, or *eager* language, i.e. function arguments are evaluated before calling a function, and expressions are computed before being stored in a variable or a class field. This is by opposition to a *lazy* language such as Haskell where expressions will generally only be computed if their value is actually needed.
- Finally, VSOP has no support for *polymorphism* (like C++ templates or Java generics), other than *subtyping* (i.e. allowing to use a child class where one of its ancestor class is expected).

The remainder of the document is organized as follows:

- Section 2 describes the lexical structure of VSOP.
- Section 3 describes the language syntax.
- Section 4 describes the semantics associated to the language, how it is typed and evaluated.
- Section 5 discusses a set of suggested extensions of the basic VSOP language.

2 Lexical Structure

This section describes how the source code character stream should be transformed into VSOP tokens. All token definitions are given in EBNF notation, assuming the following definitions:

```
lowercase-letter = "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h" | "i" | "j"
                  | "k" | "l" | "m" | "n" | "o" | "p" | "q" | "r" | "s" | "t"
                  | "u" | "v" | "w" | "x" | "y" | "z";
uppercase-letter = "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H" | "I" | "J"
                  | "K" | "L" | "M" | "N" | "O" | "P" | "Q" | "R" | "S" | "T"
                  | "U" | "V" | "W" | "X" | "Y" | "Z";
letter = lowercase-letter | uppercase-letter;
bin-digit = "0" | "1";
digit = bin-digit | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9";
hex-digit = digit | "a" | "b" | "c" | "d" | "e" | "f" | "A" | "B" | "C" | "D"
           | "E" | "F";
```

Lexical ambiguities are resolved through the longest prefix match rule. E.g. the input string "<=" is to be interpreted as token `lower-equal` rather than token `lower` followed by token `equal`, while `if42` is a valid object identifier notwithstanding the fact that it begins with reserved keyword `if`.

2.1 Whitespace

The following characters are considered whitespace: space, horizontal tabulation, line feed, carriage return and form feed (character to hint a page break).

```
whitespace = { " " | tab | lf | ff | cr };
tab = ? ASCII character 9 ?;
```

```
lf = ? ASCII character 10 ?;
ff = ? ASCII character 12 ?;
cr = ? ASCII character 13 ?;
```

Whitespace characters are generally ignored, but do separate adjacent tokens that would otherwise make a single token. *E.g.* `someFun42` is a single identifier while `someFun 42` gives two tokens, the identifier `someFun` followed by the integer literal `42`.

While line feeds are ignored (*i.e.* they are not returned as part of a token), they still play a role in single-line comments and character strings, as explained in sections 2.2 and 2.7.

2.2 Comments

Single-line comments are introduced by `//` and continue up to the next line feed.

Multiple-line comments are introduced by `(*` and continue up to the corresponding `*)`. Contrarily to most C-derived languages, where comments do not nest, these comments may be *nested* arbitrarily. The following code is valid, and contains only a comment:

```
(* Here is a comment. (* Valid nested comment. *) Still commented. *)
```

A multi-line comments should be explicitly terminated, so that reaching end-of-file while inside such a comment is an error in VSOP. However, one can end a single-line comment with end-of-file instead of a line feed.

2.3 Integer Literals

An integer literal is a sequence of one or more digits, optionally prefixed by `0x` for hexadecimal numbers. The default radix when no prefix is used is 10.

```
integer-literal = digit { digit } (* base-10 number *)
                  | "0x" hex-digit { hex-digit }; (* base-16 number *)
```

Rather than interpret a number such as `0xabcdefgh` as an integer literal `0xabcdef` followed by object identifier `gh`, it is a lexical error in VSOP. An integer literal followed by an identifier should use some whitespace to separate the two tokens. The same holds true for decimal numbers.

2.4 Keywords

The list of keywords in the VSOP language is given in listing 1. Keywords are case-sensitive.

<code>and</code>	<code>extends</code>	<code>isnull</code>	<code>string</code>
<code>bool</code>	<code>false</code>	<code>let</code>	<code>then</code>
<code>class</code>	<code>if</code>	<code>new</code>	<code>true</code>
<code>do</code>	<code>in</code>	<code>not</code>	<code>unit</code>
<code>else</code>	<code>int32</code>	<code>self</code>	<code>while</code>

Listing 1: VSOP reserved keywords.

2.5 Type Identifiers

In VSOP, user-defined types always begin with an uppercase letter, and can contain letters, digits and underscores.

```
type-identifier = uppercase-letter { letter | digit | "_" };
```

Primitive built-in types like `bool` begin with a lowercase letter, but are keywords.

2.6 Object Identifiers

In VSOP, all identifiers that are neither keywords, nor type names denotes *objects*. Those identifiers begin with a lowercase letter, and can contain letters, digits and underscores.

```
object-identifier = lowercase-letter { letter | digit | "_" };
```

2.7 String Literals

Strings are sequences of characters enclosed inside a pair of double-quotes ("). A string cannot contain a literal line feed, or the end-of-file. All other characters are interpreted literally, with the exception of the backslash (\) which introduces a character escape sequence. All possible character escape sequences are

```
\b    backspace
\t    horizontal tabulation
\n    line feed
\r    carriage return
\"    double-quote (not ending the string)
\\    backslash
\xhh  character with byte value hh in hexadecimal.
```

A \ not followed by one of the above escape sequences is an error in VSOP, unless it occurs at the end of a line. A backslash directly followed by a new line and, optionally, a number of spaces and/or tabulations will be ignored, so that

```
"A supposedly very very long str\
  ing."
```

is strictly equivalent to "A supposedly very very long string.". This allows to break strings to make them more legible, or enforce some column limit on the source code.

Comments do not occur inside strings, so that the string "Here comes (* Zorglub *)" actually contains the word Zorglub, as do the string "Uninterrupted string // Zorglub".

All strings should be explicitly terminated, so that reaching end-of-file while inside a string is an error in VSOP.

```
string-literal = '"' { regular-char | escaped-char } '"';
regular-char  = ? any valid, non-escaped character as described above ?;
escaped-char  = "\" escape-sequence;
escape-sequence = "b" | "t" | "n" | "r" | "'" | "\"
                  | "x" hex-digit hex-digit
                  | lf { " " | tab }; (* ignored, along with previous "\" *)
```

2.8 Operators

Here is the list of VSOP operators:

lbrace = "{";	comma = ",";	dot = ".";
rbrace = "}";	plus = "+";	equal = "=";
lpar = "(";	minus = "-";	lower = "<";
rpar = ")";	times = "*";	lower-equal = "<=";
colon = ":";	div = "/";	assign = "<-";
semicolon = ";";	pow = "^";	

3 Syntax

The syntax of VSOP is given by the grammar in listing 2. As given, this grammar is ambiguous. It is disambiguated with the precedence and associativity rules in table 1, and the following rules:

- In *if-then-else*, *while*, and *let* expressions, the embedded expressions are taken to be as long as possibly allowed by the grammar. *E.g.*

```
1 + if cond then 0 else 40 + 2
```

is equivalent to

```
1 + (if cond then 0 else (40 + 2))
```

and not

```
1 + (if cond then 0 else 40) + 2
```

- An *else* branch should be associated with closest *if-then*. *E.g.*

```
if cond1 then if cond2 then doThis() else doThat()
```

should be interpreted as

```
if cond1 then { if cond2 then doThis() else doThat() }
```

and not as

```
if cond1 then { if cond2 then doThis() }  
              else { doThat() }
```

Operator	Precedence	Associativity
.	1	left
^	2	right
unary -	3	right
isnull	3	right
*	4	left
/	4	left
+	5	left
-	5	left
<	6	non-associative
<=	6	non-associative
=	6	non-associative
not	7	right
and	8	left
<-	9	right

Table 1: Precedence and associativity rules.

4 Semantics

This section gives the *scoping rules* (*i.e.* describing which entity a name is referring to), the *typing rules* and the *evaluation rules* of VSOP.

VSOP is a *statically typed* language. The compiler typechecks the program at compile time and ensures that no type error can occur at runtime. It does so by assigning a type to every expression, according to the type annotations provided by the developer and the language typing rules.

```

program = class { class };
class = "class" type-identifier [ "extends" type-identifier ] class-body;
class-body = "{" { field | method } "}";
field = object-identifier ":" type [ "<-" expr ] ";";
method = object-identifier "(" formals ")" ":" type block;
type = type-identifier | "int32" | "bool" | "string" | "unit";
formals = [ formal { "," formal } ];
formal = object-identifier ":" type;
block = "{" expr { ";" expr } "}";
expr = "if" expr "then" expr [ "else" expr ]
      | "while" expr "do" expr
      | "let" object-identifier ":" type [ "<-" expr ] "in" expr
      | object-identifier "<-" expr
      | "not" expr
      | expr "and" expr
      | expr ("=" | "<" | "<=") expr
      | expr ("+" | "-") expr
      | expr ("*" | "/" ) expr
      | expr "^" expr
      | "-" expr
      | "isnull" expr
      | object-identifier "(" args ")"
      | expr "." object-identifier "(" args ")"
      | "new" type-identifier
      | object-identifier
      | "self"
      | literal
      | "(" ")"
      | "(" expr ")"
      | block;
args = [ expr { "," expr } ];
literal = integer-literal | string-literal | boolean-literal;
boolean-literal = "true" | "false";

```

Listing 2: EBNF grammar describing VSOP syntax.

4.1 Classes

All code in VSOP is contained inside classes. All class names also define a type and are globally visible. I.e. they can be used anywhere in the input source file, including before the class definition, or inside the class definition itself. Classes cannot be redefined.

A class definition consists of lists of *fields* and *methods*, both of which can be empty. A field of a class C specifies a variable that is part of the state of *objects* (also known as *instances*) of class C. A method of class C is a procedure that manipulates the variables of objects of class C.

Neither a field nor a method can be defined multiple times in the same class (even with different types). However, a field and a method of a class can have the same name (they reside in different *name spaces*).

As you have seen in section 3, the types of fields, as well as the types of formal arguments and return types of methods must be given explicitly by the developer.

New

A new object of a class is created by using the `new` operator. The expression `new C` creates a fresh object of class C. The object can be thought of as a record containing a space for each of the fields of class C, along with a pointer to a table of its methods. The `new` operator will not only allocate space for the class instance, but will also initialize its fields (and methods table pointer), as explained below.

The type of expression `new C` is C.

There is no corresponding `delete` operator in VSOP, which sports *automatic memory management*. Objects which cannot be used anymore will be reclaimed by a *garbage collector*.

Inheritance

If a class C *extends* a class P, it inherits all the fields and methods of class P in addition to its own fields and methods. Class C is called a *child* of class P. Class P is called a *parent* of class C.

The inheritance relation is *transitive*. If a class C extends a class B, which itself extends a class A, then class C also extends class A.

A *child class* can be used in any place where one of its parent class can be used. We say that the type of the child class *conforms* to the type of the parent class.

It is illegal to redefine a field of a parent class in a child class. However, a child class can *override* a method of a parent class. The redefined method must have the exact same type (i.e. same arguments and return type) than in the parent class, and takes precedence over it (i.e. *object-oriented dispatch on object of the child class will call the child method*).

VSOP only supports *single inheritance*, i.e. a class can only extend a single parent class. Moreover, there can be no cycles in the inheritance relation, i.e. a class may not extend one of its child class as in the following invalid example:

```
class Bogus extends OtherBogus { (* ... *) }
class OtherBogus extends Bogus { (* ... *) }
```

It is important to distinguish between the *static* type of an expression which is inferred by the compiler at compile time, and the *dynamic* type of that expression during execution. This distinction is necessary because it is not in general possible for the compiler to infer the exact type of values at runtime, due to inheritance (and conformance). The compiler will however ensure that static types are *sound* with respect to all possible dynamic types. See listing 3 for an example illustrating the difference between static and dynamic types.

?

When not otherwise specified, a type will refer to the static type in the remainder of this document.

Dispatch

The dot operator allows to call methods of an object through *object-oriented dispatch*. E.g. the code `myObject.someMethod(arg1, 42)`

calls the `someMethod` method of the class of object `myObject` with arguments `arg1` and `42`. There could be several methods `someMethod` in different classes. Which `someMethod` is called depends on the **dynamic** type of object `myObject`.

```
class P { name() : string { "P" } }
class C extends P {
  name() : string { "C" }
  onlyInC() : int32 { (* ... *) }
}
class Other {
  myMethod() : string {
    let p : P <-           // Declared type is P => static type is P.
      if inputInt32() = 0   // inputInt32() will ask the user for a number.
      then new C           // `new C` valid here as C conforms to P.
      else new P
    in {
      p.onlyInC(); // Type error. Static type is P, not C. Would be valid
                  // if the user typed 0, but we cannot tell at compile
                  // time.
      p.name() // Dispatch is done using dynamic type.
               // Will return "P" or "C" depending on what the user typed.
    }
  }
}
```

Listing 3: Static types vs dynamic types.

Fields

If a field has the optional initializer, it will be executed when a new object is created, and the resulting value assigned to the field. The type of the initializer must conform to the (static) type of the field. The object's fields and methods are not yet in scope in the field initializer, as the object is not initialized yet. The object itself (`self`) is not available either.

If no initializer is present, the field is *default-initialized*. Values of type `int32` will be set to `0`, values of type `bool` will be set to `false`, and values of type `string` will be set to `"`". Fields of class types will be set to the special value `null`.

The `null` value is similar to `NULL` in C or `null` in Java. It can be used wherever an object can be used (e.g. passed as an argument, stored into a variable, etc.), but an attempt to dispatch (i.e. call a method) on a `null` value will result in a runtime error. Note that there is no explicit name for `null` values in VSOP. One can only create a `null` value by default-initializing an object variable or field.

One can test whether or not a value is `null` using the `isNull` operator.

Fields are initialized in the order of inheritance, beginning with `Object` down to the object class. Inside a class, fields are initialized in the same order as they appear in the class.

Fields are local to the class they are declared in, and to child classes. They are thus *protected* and can only be manipulated through the class methods, or child class methods.

Methods

All methods in basic VSOP have global scope, they are *public*¹.

A method can have zero or more formal parameters. The identifiers used in the formal parameters list must be distinct.

The result of the method invocation is the result of the evaluation of its body. Within the method body, field names refer to the corresponding fields of the object upon which the method is called. Formal parameter names refer to the corresponding arguments. If a parameter name has the same name than a field, it takes precedence (i.e. it hides the corresponding field). The special variable *self* refers to the object itself.

The type of the method body must conform to its declared return type.

The Object Class

The pre-defined Object class is the default parent class of a class, when its class definition does not explicitly extends a class. As VSOP only support acyclic, single inheritance, it follows that all classes are members of a tree rooted at Object, which is the common ancestor of all classes.

The pre-defined Object class enables (basic) input/output in VSOP. It has the following prototype:

```
class Object {
  print(s : string) : Object { (* print s on stdout, then return self *) }
  printBool(b : bool) : Object { (* print b on stdout, then return self *) }
  printInt32(i : int32) : Object { (* print i on stdout, then return self *) }
  inputLine() : string {
    (* read one line from stdin, return "" in case of error *) }
  inputBool() : bool {
    (* read one boolean value from stdin, exit with error message in case of
       error *) }
  inputInt32() : int32 {
    (* read one integer from stdin, exit with error message in case of
       error *) }
}
```

The Main Class

A valid VSOP program must provide a Main class, with a method *main* with no arguments and returning *int32*. This serves as an entry point to the program.

When the program is run, an object of class Main is created, and its method *main* is called. The return value of the method is used as the program exit code².

4.2 Expressions

Literals

Literal constants are the simplest expressions. Each literal constant evaluates to its value, and has the following type:

¹Which is not very good for encapsulation and information hiding. See 5.14 if you want to change that.

²As specified by the POSIX standard, the exit code should be 0 if the program executed successfully, and some error code different from 0 if an error occurred.

- `true` and `false` are of type `bool`.
- `integer literals` have type `int32`, and can represent 32-bit signed integers.
- `string literals` have type `string`. A string constant is a contiguous sequence of bytes corresponding to the string characters, followed by a NUL character.
- The only inhabitant of the `unit` type is `()` (also called unit). `unit` is used where `void` would be used in C, but has one valid value `()`, which can be passed as argument, stored into a variable, etc. The actual representation of unit values does not matter, as the type itself contains all the information. E.g. say you expect an argument of type `unit`. As VSOP is statically typed, you don't need to actually read the argument value at runtime, you already know it can only be `()`, the only inhabitant of `unit`. You can think of the unit value as an empty tuple, hence the `()` notation.

Identifiers

The names of local variables (introduced by `let ... in`, see below), formal parameters of methods, `self` and class fields are expressions. They evaluate to the current value associated with the local variable, parameter or field, and have the corresponding type.

The *binding* of an identifier references the innermost lexical scope that contains a declaration for that identifier, or to the field of the same name if there is no other declaration. A field can thus be hidden by a formal parameter or local variable, a formal parameter can be hidden by a local variable, and a local variable can be hidden by another local variable declaration with the same name within its scope.

The exception to the previous rule is the identifier `self`, which is implicitly bound to the current object in every method, and cannot be hidden (it is a syntax error to declare a field, formal parameter or local variable named `self`). Note that `self` is only bound in method bodies, it is not in scope in field initializers.

Assignments

An assignment of the form `<id> <- <expr>` first evaluates `<expr>`, then assign its value to identifier `<id>`. The type of `<expr>` must conform to the declared type of `<id>`. The resulting value of the whole assignment is the value of `<expr>`.

Dispatch

Object-oriented dispatch was introduced in section 4.1. More precisely now, consider an expression of the form

```
<expr_0>.<id>(<expr_1>, ..., <expr_n>)
```

To typecheck the dispatch, assuming the **static** type of `<expr_0>` is `P`, the compiler will check that class `P` has (or inherits) a method `<id>` with `<n>` formal parameters (`<n>` can be zero), such that the static type of the *i*-th actual argument `<expr_i>` conforms to the type of the *i*-th formal parameter.

To evaluate the dispatch, `<expr_0>` is evaluated first. Arguments `<expr_1>`, ..., `<expr_n>` are then evaluated from left to right. Finally, assuming `<expr_0>` has **dynamic** type `C`, the method `<id>` of class `C` is invoked with `self` bound to the value of `<expr_0>` and its formal parameters bound to the values of the actual arguments `<expr_1>`, ..., `<expr_n>`. The value of the expression is the value returned by the method invocation.

A second form of dispatch

```
<id>(<expr_1>, ..., <expr_n>)
```

is simply a shortcut for a dispatch to `self`. It is entirely equivalent to

```
self.<id>(<expr_1>, ..., <expr_n>)
```

Conditionals

In a conditional of the form

```
if <cond> then <expr_t> else <expr_e>
```

the condition `<cond>` must be of `bool` type.

The types of both branches `<expr_t>` and `<expr_e>` must agree, which we define as follows:

- If both branches have class type, the types agree and the resulting type of the conditional is the class of the first common ancestor of the two branches.
- If both branches have another primitive type, the types agree if and only if there are the same. The resulting type of the conditional is the type of both branches.
- If (at least) one branch has type `unit`, the types agree and the resulting type of the conditional is `unit`.
- Else, the type of both branches don't agree, and it is a typing error.

The evaluation proceeds as follows. The condition `<cond>` is evaluated first. If its value is `true` the expression `<expr_t>` is evaluated and `<expr_e>` is ignored. If the condition value is `false`, `<expr_e>` is evaluated and `<expr_t>` is ignored. The resulting value of the conditional is `()` if the conditional has type `unit`, and the value of the chosen branch otherwise.

A conditional of the form

```
if <cond> then <expr_t>
```

without an `else` branch is just a shortcut for

```
if <cond> then <expr_t> else ()
```

Loops

In a loop of the form

```
while <cond> do <expr>
```

the condition `<cond>` must have type `bool`. The type of `<expr>` can be any type. The type of the loop is `unit`³.

The condition is evaluated before each iteration of the loop. If the condition value is `false`, the loop terminates and `()` is returned. If the predicate is `true`, the body of the loop is evaluated and the process repeats.

Blocks

Expressions in a block are evaluated in the same order as they appear in the block. They can have any type.

The resulting type and value of the whole block are those of its last expression.

³If you wanted to use the type of `<expr>`, think about the case where the condition is false from the start.

Let

A local variable declaration has the form

```
let <id> : <type> [<- <init_expr>] in <body_expr>
```

If an initializer `<init_expr>` is provided, it is evaluated first and its resulting value is bound to `<id>` in the body. The type of the initializer must of course conform to the declared type `<type>`. If no initializer is provided, `<id>` is bound to the default initializer of `<type>` (as explained in section 4.1) in the body. The body is then evaluated, and returned as the value of the whole `let` expression. The type of the `let` expression is the type of the body.

Note that the scope of the binding is just `<body_expr>`. After the `let` expression, `<id>` takes back its previous binding if it had one (else, it becomes undeclared again).

Arithmetic, Logic and Comparison Operations

All binary operators except `and` first evaluate their left-hand side operand, then their right-hand side operand.

Arithmetic operations, `<` and `<=` are only defined on operands of type `int32`. Arithmetic operations return a value of type `int32` (VSOP only supports integer division), according to the usual semantics. `<` and `<=` also follow usual semantics, returning a `bool`.

The equality comparison operator `=` is special. It can be used on any two values with the same primitive type, returning `true` if the values are the same and `false` if they are different. It can also be used between any two class-type objects (not necessarily with the same class). **It returns `true` if both objects have the same memory address, and `false` otherwise.** It is a typing error to try to compare two values with different primitive types, or a value with a primitive type and a value with a class type.

Logical operators act on `bool` values, and also return a `bool`, according to the usual semantics.

`and` is *short-circuiting*, i.e. it first evaluates its left-hand side, and only if it is `true` will it evaluate its right-hand side. The right-hand side must still be a valid expression, however (i.e. it must be semantically correct, and typecheck to `bool`).

5 Suggested Extensions

The VSOP could be extended and/or changed in an infinite number of ways. You could improve the syntax, add more functionality, ... Here are some suggestions of extensions. This list is far from being exhaustive, inspire yourself from other programming languages to find many other potential improvements.

5.1 Add New Control-Flow Structures

With only *if-then-else* conditionals and *while* loops, basic VSOP is not always very convenient to program with. Extend it with other control-flow structures such as *do-while*, *for*, *switch*, ...

5.2 Indentation-based Syntax

VSOP syntax can be quite misleading. E.g. the following code

```
if cond then
  expr1;
  expr2
```

looks like `expr2` will only be evaluated when `cond` is `true`, while it will actually always be evaluated (it is outside the conditional).

Modify the VSOP language so that blocks are defined by indentation like in Python, rather than relying on curly braces.

5.3 Easier Let-In Constructs

The syntax of *let-in* constructs in VSOP is easy to parse (and check), but quite painful to use, e.g. code such as

```
let x : ... <- ... in {
  ...;
  let y : ... <- ... in {
    ...;
    let z : ... <- ... in {
      ...
    }
  }
}
```

is ugly and occurs quite frequently.

Modify the VSOP language in order to make the scope of a *let-in* construct extends as far as possible in the current block, such that the previous code can be rewritten simply as

```
let x : ... <- ... in
...;
let y : ... <- ... in
...;
let z : ... <- ... in
... // x, y and z all in scope here
```

5.4 Add other numeric types

Having only 32-bit integers is quite limiting. One could add other numeric types such as `double` to VSOP. Once done, you could also consider adding some ways to convert (automatically or not) between different numeric types.

5.5 Add Missing Operators

Basic VSOP only supports a very limited subset of usual operators (e.g. it has `and`, but not `or`; it has `<`, but not `>`). Add missing operators.

5.6 Add a Foreign Function Interface (FFI)

With very few built-in I/O functions (in class `Object`), VSOP does not give much access to the system (e.g. it does not allow to open sockets, program the OpenGL pipeline, ...), and offers little help for development.

You could extend the standard library to provide additional functionalities, but there will always be some functionalities which are provided by some external libraries, but not by the VSOP standard library. Moreover, it would be quite tiresome to have to rewrite every useful library in VSOP, rather than reuse existing libraries.

Instead, one could add a *foreign function interface* (FFI) to VSOP. A foreign function interface allows to call library functions written in another language (most often C) from your language.

E.g., you could add an external keyword which allows to declare library functions, as in the following example:

```
external strlen(s : string) : int32; // Declare C's strlen function
                                     // Here we assume string has the same
                                     // representation as in C

class String {
  s : string;
  init(theS) : String { s <- theS }
  length() : int32 {
    strlen(s) // No method strlen on self, but external strlen is defined
  }
}
```

5.7 Dispatch on Parent Class

When a method is overridden in VSOP, it completely hides the parent method. However, it is often useful to be able to call a parent method from an overridden child method. Modify VSOP so that you can make a dispatch on a specific class. E.g. add a construct

```
<expr>::<type>.<id>(<expr1>, ..., <exprN>)
```

such that the method called is the one of <type>, rather than the one of the dynamic type of the expression. Of course, the type of <expr> should conform to <type>.

Here is an example use:

```
class Point2D {
  x : int32;
  y : int32;

  printPoint() : Object { print("x=").printInt(x).print(", y=").printInt(y) }
}
class Point3D extends Point2D {
  z : int32;

  printPoint() : Object {
    self::Point2D.printPoint();
    print(", z=").printInt(z)
  }
}
```

5.8 Addition of Arrays

Many useful algorithms critically depend on arrays (*i.e.* a container data structure where getting or setting the value at any index can be done in $\mathcal{O}(1)$ time).

VSOP already allows the definition of container classes such as linked lists, or various tree-based data structures, but does not allow the creation of arrays (the language is not expressive enough).

Add support for arrays in VSOP. These arrays could be specific to one type (e.g. adding an `IntArray` built-in class), but should ideally be *parametric* in the element type, so that you can declare arrays of any type (e.g. `array<int32>`, `array<MyClass>`, `array<array<bool>>`, ...).

5.9 Local Type Inference

Most developers don't like to have to explicitly give types to every definitions, when the type can be trivially inferred by the compiler itself. *E.g.* in

```
let mc : MyClass <- new MyClass in ...
```

it is quite obvious than `mc` has the type `MyClass`. The type annotation does not bring much, and can even make the code harder to read.

Modify VSOP so that the type of `let` bindings becomes optional. If no type is provided, just use the (static) type of the initializer as the type for the defined identifier.

The return type of a method can also be inferred quite easily. However, doing full type inference (*i.e.* requiring no type annotations from the developer, as is done in Haskell or OCaml), is more complex.

5.10 Top-level Functions and/or Static Methods

It is a bit artificial to require that every dispatch should be on an object instance. *E.g.* the `print` method of the `Object` class never use its implicit `self` first argument. It could as well be a top-level *function* (rather than a method), or a *static* method (*i.e.* a method that can be called directly on a class rather than an object, *e.g.* using `Object::print(s)`).

Extend VSOP so that it also allows the definition of functions, and/or add a `static` keyword that transform an object method into a static (*i.e.* class) method.

5.11 Add Static Fields

Similarly to the static methods discussed above, it might be interesting to add support for *static fields*, *i.e.* fields that have one single instance shared by all objects of the class, rather than one copy per object instance.

5.12 Multiple Inheritance or Interfaces

Although it comes with its own set of problems, *multiple inheritance* can be nice at times. Extend VSOP so that a class can inherit multiple parents. This not only requires implementation in the compiler, but also careful language design to avoid most multiple-inheritance pitfalls (like the diamond problem). *E.g.* contrast the approach followed by C++ with Scala mixin class composition.

Alternatively, you could also allow only the multiple inheritance of *interfaces*, like in Java.

5.13 Generics/Templates

Having to define an `Int32List` for lists of `int32`, a `StringList` for lists of `string`, *etc.* is painful and violates the DRY principle.

Instead, it would be nice if VSOP allowed to define a `List<T>` class, parameterized on element type `T` (see similar discussion for arrays above).

Add supports for parametric classes in VSOP.

5.14 Visibility

In basic VSOP, all fields are always *protected*, while all methods are necessarily *public*. This is not very flexible. Modify the language so that the developer can choose the visibility of fields or methods between *private*, *public* and *protected*.

Try to choose defaults that encourage information hiding and encapsulation.

5.15 Default and Named Arguments

Languages where you can give default values to method arguments (which then become optional for the callers) are syntactically nice, *e.g.*

```
join(ss : StringList, sep = " ") : string { (* ... *) }
...
  join(xs);
  join(ys, ", ")
```

is nicer than

```
join(ss : StringList) : string { joinWithSeparator(ss, " ") }
joinWithSeparator(ss : StringList, sep : string) { (* ... *) }
...
  join(xs);
  joinWithSeparator(ys, ", ")
```

Named arguments also help readability and safety when a method has many arguments with the same types, *e.g.*

```
(* No need to remember arguments order *)
drawRectangle(x => 0, width => 100, height => 50, y => 10)
```

5.16 Add Support for Multiple Return Values (Anonymous Structs)

It is quite inconvenient to only be able to return a single type. If a method needs to return a pair of objects of type A and B, the developer needs to first define a class with fields (and getters/setters) for the values of A and B.

Modify VSOP so that one method can return a list of values, *e.g.*

```
myMethod() : (A, B) { (* ... *) (a, b) }
...
  let (foo, bar) : (A, B) <- myMethod() in ...
```

You could restrict such a scheme for method return type, or allow it everywhere (like { %A, %B } in LLVM, or a (a, b) tuple in OCaml).

5.17 Add an Error Handling Mechanism

Errors are difficult to handle in basic VSOP. You cannot just print an error message and exit (there is no pre-defined exit function), you cannot throw an exception, you cannot return an error flag in addition to the return value, *etc.*

Add a mechanism to ease error handling in VSOP. *E.g.* you could do one of the following:

- Add an exception mechanism (*e.g.* based on `setjmp/longjmp`);
- Encapsulate return values into parametric types, *e.g.*

```
someMethod() : Result<int32> {
  (* ... *)
  if hadError then (new Result).makeError("reason for the error")
  else (new Result).makeResult(theValue)
}
...
```

```

let x : Result<int32> <- someMethod() in
if x.isValid() then
  x.value() + 42 // use the stored value
else
  (* Handle error ... *)

```

- Allow to pass some primitive value *by reference*, so that they can be set in the callee and the change is reflected in the caller, e.g.

```

someMethod(a : int32, b : int32, isError : ref bool) {
  (* can set isError for caller here *) }

```

5.18 Separate Compilation

It is not very practical that all of a VSOP program must lie in a single file. Modify the VSOP compiler and language to allow the fragmentation of VSOP programs over multiple files, which can then be compiled separately, before being *linked* together. You can add additional restrictions (e.g. imposing one file per class) if you like.

5.19 Garbage Collection

This is not a language extension *per se*, but VSOP is supposed to be garbage-collected. Modify your compiler so that memory is automatically reclaimed when objects are no longer in use. You can experiment with your own garbage collection scheme (might be quite time-consuming), or reuse an existing garbage collection library such as the BOEHM-DEMERS-WEISER libgc.

A Examples

These examples are here just to give you a feel of the VSOP language. They are not normative, and might change a little by the end of the course.

A.1 Factorial

```
class Main {
  factorial(n : int32) : int32 {
    if n < 2 then 1
    else n * factorial(n - 1)
  }

  main() : int32 {
    print("Enter an integer greater-than or equal to 0: ");
    let n : int32 <- inputInt32() in
    if n < 0 then {
      print("Error: number must be greater-than or equal to 0.\n");
      -1
    } else {
      print("The factorial of ").printInt32(n).print(" is ");
      printInt32(factorial(n)).print("\n");
      0
    }
  }
}
```

A.2 Linked List

```
class List {
  isNil() : bool { true }
  length() : int32 { 0 }
}

(* Nil is nothing more than a glorified alias to List *)
class Nil extends List { }

class Cons extends List {
  head : int32;
  tail : List;

  init(hd : int32, tl : List) : Cons {
    head <- hd;
    tail <- tl;
    self
  }

  head() : int32 { head }
  isNil() : bool { false }
  length() : int32 { 1 + tail.length() }
}

class Main {
```

```

main() : int32 {
    let xs : List <- (new Cons).init(0, (new Cons).init(
                                                1, (new Cons).init(
                                                    2, new Nil))) in {
        print("List has length ");
        printInt32(xs.length());
        print("\n");
    }
}

```